

Realistic Neural Modelling Using GENESIS

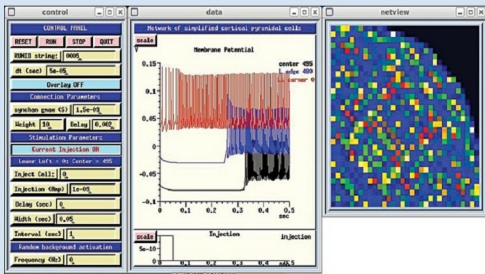
JAI SACHITH PAUL

Bio-electronic engineers around the world are working on neural simulations for quite a long time. They have been sorting out ways to make a physical description of the nervous system for developing general principles of organisation, function and computation of the same. For this, they are simulating the behaviour of various brain structures like the cerebral cortex.

All researchers are considered to be futile if these cannot contribute anything to the society. One question that disturbs the researchers in this area is, "Can we develop software that can program an implanted device for simulating the nervous system in individuals with neurodegenerative diseases?" GEneral Neural Simulation System (GENESIS) is an answer to this question. Let us take a look at this software in detail and an interesting case study at the end.

What is GENESIS

GENESIS is a simulation environment that enables the user to construct realistic models of neurobiological systems. We can simulate neural systems of any level of complexity, from sub-cellular components and biochemical reactions to whole cells, networks of cells and systems-level models using this tool. GENESIS allows the user to exchange, modify and re-use models or model components in the simulator. This is made possible with the help of the object-oriented approach



Custom scripted GUI for a network simulation in GENESIS

Realistic modelling: Inspired from the history of science

When we study the history of science, one fact becomes quite evident. For the quantitative understanding of physical systems, we should initially have a realistic model of the same. General principles, like the existence of gravitational force, were evolved from studies on realistic models. Dr James Bower and his team at Caltech labs felt that a physical description was necessary for understanding the physical structure and carrying out simulations that study the fundamental relationship between the structure of the brain and its measurable behaviour. They believed that general principles of organisation, function and computation will only emerge if this description is constructed. Since its release for general use in 1988, GENESIS provides researchers an effective framework for developing this physical description.

taken by the software and its high-level simulation language.

GENESIS is implemented in C language using X-Windows system, and runs under most varieties of UNIX, including Linux. The design of the software is such that it could easily be adapted for parallel computing. An extension to GENESIS, called parallel GENESIS (PGENESIS), runs on almost any parallel cluster, SMP, supercomputer or network of workstations, where MPI and/or PVM is supported, and on which serial GENESIS itself is run-able.

Why use a general simulator for neural modelling. The intention of using a general simulation system is clearly described by Dr Bower and Beeman in *The Book of GENESIS*. If the tool had a dedicated code for particular simulations, simulation speed and memory utilisation would have been optimal. But the processing power and speed of computers have increased over the years. The price of the memory has also decreased marginally. So, sticking to these two parameters is not an absolute necessity any more.